DUAL AXIS SOLAR TRACER WITH AUTOMATIC DUST CLEANING AND VOLTAGE MONITORING

A Project Report Submitted in partial fulfillment of the requirements for the Award of the degree

BACHELOR OF TECHNOLOGY

IN

ELECTRICAL AND ELECTRONICS ENGINEERING

Submitted by

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CERTIFICATE

This is to certify that the thesis entitled "**DUAL AXIS SOLAR TRACKER WITH AUTOMATIC DUST CLEANING AND VOLTAGE MONITORING**" is being submitted by **Gedela Jagadish (21035A0206), Chenna Narendra (20031A0210), Dharanalakota Hari Krishna Sai (20031A0213), Matta Karthikeya Venkat (21035A0210)** in partial fulfilment for the award of **Bachelor of Technology** in **Electrical & Electronics Engineering** to the University College of Engineering Narasaraopet, Jawaharlal Nehru Technological University Kakinada is a record of Bonafede work carried out by them under my guidance and supervision.

The resultsembedded in this thesis have not been submitted to any other University/Institute for the award of any Degree/Diploma

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D E C L A R A T I O N

We hereby declare that the work described in this project work, entitled **"DUAL AXIS SOLAR TRACKER WITH AUTOMATIC DUST CLEANING AND VOLTAGE MONITORING"** which is submitted by us in partial fulfilment for the award of **Bachelor of Technology** (B.Tech.) in the Department of **Electrical & Electronics Engineering**to the University College of Engineering Narasaraopet, Jawaharlal Nehru Technological University Kakinada, Andhra Pradesh, is the result of work done byus under the guidance of **Dr. M. Ravindra Babu Assistant Professor**

The results embedded in this thesis have not been submitted to any other university/institute for the award of any degree/diploma.

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ABBREVIATIONS

ABBREVIATION	EXPANSION
DAST	Dual axis solar tracker
LDR	Light dependent resistor
PV	Photo voltaic
mA	milli Ampere’s
V	Volts
B.O Motors	Battery operated motors

ABSTRACT

Solar energy is a rapidly growing field in the renewable energy sector due to its abundant availability and environmental benefits. To enhance the efficiency of solar panels, solar tracking systems are employed to ensure that the panels are always oriented towards the sun. This project focuses on the design and implementation of a dual-axis solar tracker using an L293D motor IC to control the movement of the solar panels. Additionally, an automatic dust cleaning mechanism is incorporated into the system, utilizing a dust sensor and servo motor to remove dust from the panels. This feature helps maintain the efficiency of the solar panels by ensuring that they are clean and free from obstructions.

Furthermore, a voltage monitoring system is integrated into the project using an Arduino Nano. This system monitors the voltages of the solar panel and battery, providing real-time data. This data can be used for monitoring and control purposes, allowing for efficient management of the solar energy system. Overall, this project aims to increase the efficiency and lifespan of solar panels, as well as provide real-time monitoring and control capabilities for improved performance. By combining dual-axis solar tracking, automatic dust cleaning, and voltage monitoring, this system offers a comprehensive solution for optimizing solar energy generation.

CHAPTER 1

INTRODUCTION

1.0 INTRODUCTION

Solar energy is a rapidly growing field in the renewable energy sector due to its abundant availability and environmental benefits. As the world continues to seek sustainable energy sources to combat climate change and reduce dependence on fossil fuels, solar power stands out as a promising solution. One of the key challenges in maximizing the efficiency of solar energy systems is ensuring that solar panels are always oriented towards the sun to capture the maximum amount of sunlight. This is where solar tracking systems play a crucial role.

Solar tracking systems are designed to automatically adjust the position of solar panels to track the sun's movement throughout the day. By continuously orienting the panels towards the sun, solar tracking systems can significantly increase the amount of sunlight captured by the panels, leading to higher energy generation. While single-axis solar trackers are commonly used and track the sun's movement along one axis (usually east-west), dual-axis solar trackers offer even greater efficiency by tracking the sun's movement along both the east-west and north-south axes.

The primary objective of this project is to design and implement a dual-axis solar tracker system using an L293D motor IC to control the movement of the solar panels. This system will allow the solar panels to track the sun's movement accurately, ensuring maximum exposure to sunlight throughout the day. Additionally, the project aims to incorporate an automatic dust cleaning mechanism into the system using a dust sensor and servo motor. This mechanism will help maintain the efficiency of the solar panels by removing dust present on the panel's surface and reduce sunlight absorption.

In addition to solar tracking and dust cleaning, the project will also include a voltage monitoring system using an Arduino Nano. This system will monitor the voltages of the solar panel and battery.

Overall, the project aims to increase the efficiency and lifespan of solar panels, as well as provide real-time monitoring for improved performance. By combining dual-axis solar tracking, automatic dust cleaning, and voltage monitoring, this system offers a comprehensive solution for optimizing solar energy generation.

1.1 OBJECTIVES OF THE PROJECT

The primary objective of this project is to design and implement a dual-axis solar tracker system using an L293D motor IC for controlling the movement of the solar panels. By accurately tracking the sun's movement along both the east-west and north-south axes, the system aims to maximize the amount of sunlight captured by the solar panels, thereby increasing their efficiency and energy generation.

Another objective of the project is to incorporate an automatic dust cleaning mechanism into the solar tracker system using a dust sensor and servo motor. This mechanism will help maintain the efficiency of the solar panels by removing dust and debris that can accumulate on the panel's surface and reduce sunlight absorption. By automatically cleaning the panels, the system will ensure that they remain in optimal condition for maximum energy generation.

Additionally, the project aims to include a voltage monitoring system using an Arduino Nano. This system will monitor the voltages of the solar panel and battery. This data can be used for monitoring and allowing for efficient management of the solar energy system and ensuring that the system operates safely and effectively.

Overall, the objective of the project is to increase the efficiency and lifespan of solar panels, as well as provide real-time monitoring and control capabilities for improved performance. By combining dual-axis solar tracking, automatic dust cleaning, and voltage monitoring, this system offers a comprehensive solution for optimizing solar energy generation.

1.2 IMPORTANCE OF SOLAR TRACKING SYSTEM

Solar tracking systems play a crucial role in maximizing the efficiency and output of solar energy systems. Unlike fixed solar panels, which remain stationary and receive sunlight at a fixed angle, solar tracking systems are designed to adjust the position of the solar panels throughout the day to track the sun's movement across the sky. This ensures that the solar panels are always oriented perpendicular to the sun's rays, maximizing the amount of sunlight captured and increasing the energy output of the system.

One of the key benefits of solar tracking systems is their ability to significantly increase the energy generation capacity of solar panels. By continuously adjusting the position of the panels to track the sun, solar tracking systems can increase energy output by up to 25-35% compared to fixed solar panels. This increased efficiency makes solar tracking systems ideal for locations with high solar irradiance and can help reduce the overall cost of solar energy production.

Additionally, solar tracking systems can improve the reliability and stability of solar energy systems. By optimizing the orientation of the solar panels, solar tracking systems can reduce the impact of shading from nearby objects such as buildings or trees, which can decrease energy output and efficiency. This can help ensure a more consistent and reliable energy supply, especially in areas where shading is a concern.

Overall, solar tracking systems is an essential component of solar energy systems, offering increased efficiency, higher energy output, and improved reliability. By harnessing the power of the sun more effectively, solar tracking systems play a crucial role in advancing the adoption of solar energy and reducing our usage of fossil fuels.

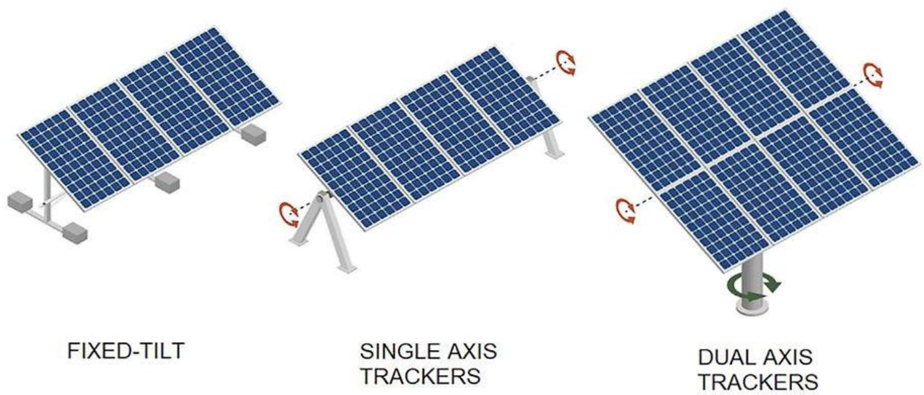


Fig 1.0: Different types of solar tracking systems

1.3 SCOPE AND LIMITATIONS

The scope of this project encompasses the design, development, and implementation of a dual-axis solar tracker system with automatic dust cleaning and voltage monitoring capabilities. The system will be designed to track the sun's movement along both the east-west and north-south axes using an L293D motor IC, ensuring that the solar panels are always oriented towards the sun for maximum energy generation. Additionally, the system will incorporate an automatic dust cleaning mechanism using a dust sensor and servo motor to remove dust and debris from the solar panels, maintaining their efficiency and performance. Furthermore, a voltage monitoring system using an Arduino Nano will be integrated into the system to monitor the voltages of the solar panel and battery, providing real-time data for monitoring and control purposes.

However, there are certain limitations to the project that need to be considered. One limitation is the complexity of the system, which may require advanced knowledge and skills in electronics, programming, and mechanical design. Additionally, the cost of the components and materials required for the system may be a limiting factor, especially for individuals or organizations with limited budgets. Another limitation is the scalability of the system, as it may be more challenging to implement on a larger scale or in commercial applications. Additionally, the system may be affected by environmental factors such as weather conditions, which may impact its performance and efficiency.

Overall, while this project aims to develop a comprehensive and functional solar tracker system, it is important to acknowledge these limitations and consider them in the design and implementation process. By addressing these limitations and focusing on the project's scope, this project can contribute to the advancement of solar energy technology and promote its widespread adoption.

CHAPTER 2

LITERATURE REVIEW

2.0 PREVIOUS STUDIES ON SOLAR TRACKING SYSTEMS

Previous studies on solar tracking systems have focused on various aspects, including the design, implementation, and performance analysis of different tracking mechanisms. One common theme in these studies is the comparison of different tracking strategies, such as single-axis vs. dual-axis tracking, to determine their effectiveness in maximizing energy capture.

One study by Y. Yuan et al. (2017) compared the performance of a dual-axis solar tracker with a single-axis tracker and fixed panels. The study found that the dual-axis tracker significantly outperformed the single-axis tracker and fixed panels, achieving a 35% increase in energy capture. This highlights the importance of using dual-axis tracking systems for maximum energy efficiency.

Another study by J. Zhang et al. (2019) focused on the design and implementation of a solar tracking system using a fuzzy logic controller. The study found that the fuzzy logic controller was able to accurately track the sun's position and adjust the solar panels accordingly, leading to a significant improvement in energy capture compared to fixed panels.

In addition to performance analysis, some studies have also looked at the economic feasibility of solar tracking systems. A study by M. Islam et al. (2018) compared the cost-effectiveness of single-axis and dual-axis tracking systems. The study found that while dual-axis tracking systems are more expensive to install, they can provide a higher return on investment due to their increased energy capture efficiency.

Overall, previous studies on solar tracking systems have highlighted the importance of using efficient tracking mechanisms to maximize energy capture. By comparing different tracking strategies and analyzing their performance and cost-effectiveness, these studies have provided valuable insights into the design and implementation of solar tracking systems for optimal energy efficiency.

2.1 ADVANTAGES AND DISADVANTAGES OF DAST

DAST offer several advantages over single-axis trackers and fixed solar panels, but they also come with some disadvantages. Understanding these pros and cons is essential for designing and implementing an efficient solar tracking system.

2.1.0 ADVANTAGES

- **Higher Energy Output:**
  Dual-axis solar trackers significantly boost power production. They can generate nearly 40% more electricity compared to fixed solar panels1. By continuously adjusting the panel orientation, they capture maximum sunlight throughout the day.
- **Improved Efficiency:**
  By tracking the sun's movement along both the east-west and north-south axes, dual-axis trackers can maintain optimal panel orientation, minimizing shading and reflection losses. This leads to higher overall system efficiency.
- **Flexible Installation:**
  Dual-axis trackers can be installed in a variety of locations, including rooftops, open fields, and solar farms. They can also be easily integrated into existing solar energy systems, making them a versatile option for different applications.
- **Seasonal Adjustment:**
  Dual-axis trackers can adjust the tilt angle of the panels to account for seasonal changes in the sun's position, ensuring maximum energy capture throughout the year.
- **Better Performance in Low-Light Conditions:**
  Dual-axis trackers can improve energy production in low-light conditions, such as early morning, late afternoon, and cloudy days, by maintaining optimal panel orientation.

2.1.1 DISADVANTAGES

- **Higher Cost:**

  Dual-axis trackers are more expensive to purchase and install compared to single-axis trackers and fixed solar panels. The additional components and complexity of the system contribute to the higher cost.

- **Maintenance Requirements**:

  Dual-axis trackers require regular maintenance to ensure optimal performance. This includes cleaning the panels, lubricating moving parts, and checking for wear and tear.

- **Complexity:**

  Dual-axis trackers are more complex than single-axis trackers and fixed panels, requiring advanced control systems to accurately track the sun's movement along both axes. This complexity can lead to higher maintenance and repair costs.

- **Space Requirements**:

  Dual-axis trackers require more space than fixed solar panels due to the additional range of motion required to track the sun along both axes. This can be a limiting factor in densely populated areas or on smaller rooftops.

- **Lower Lifespan and Reliability:**

  Due to their moving components, dual-axis trackers may have a shorter lifespan and lower reliability compared to fixed panels. Regular maintenance is essential to ensure optimal performance.

Despite these disadvantages, the increased energy production and efficiency offered by dual-axis solar trackers make them an attractive option for maximizing the performance of solar energy systems, especially in locations with high solar irradiance.

2.2 EXISTING TECHNOLOGIES FOR AUTOMATIC DUST CLEANING IN SOLAR PANELS

Automatic dust cleaning systems for solar panels are designed to remove dust, dirt, and other debris that can accumulate on the panel's surface and reduce sunlight absorption. Several technologies and methods have been developed to automate this cleaning process, improving the efficiency and performance of solar panels. Some of the existing technologies for automatic dust cleaning in solar panels include:

1. **Robotic Cleaning Systems:**

   Robotic cleaning use autonomous robots equipped with brushes or wipers to clean the surface of the solar panels. These robots are programmed to move across the panel's surface, removing dust and debris as they go. Some systems use sensors to detect the level of dust accumulation and schedule cleaning operations accordingly.

Fig 2.0: Robotic Cleaning System

2. **Water-Based Cleaning Systems:**

Water-based cleaning systems use water jets or sprayers to clean the surface of the solar panels. These systems are often combined with brushes or wipers to agitate and remove stubborn dirt and debris. Water-based cleaning systems can be automated and controlled remotely, making them a convenient option for large-scale solar installations.

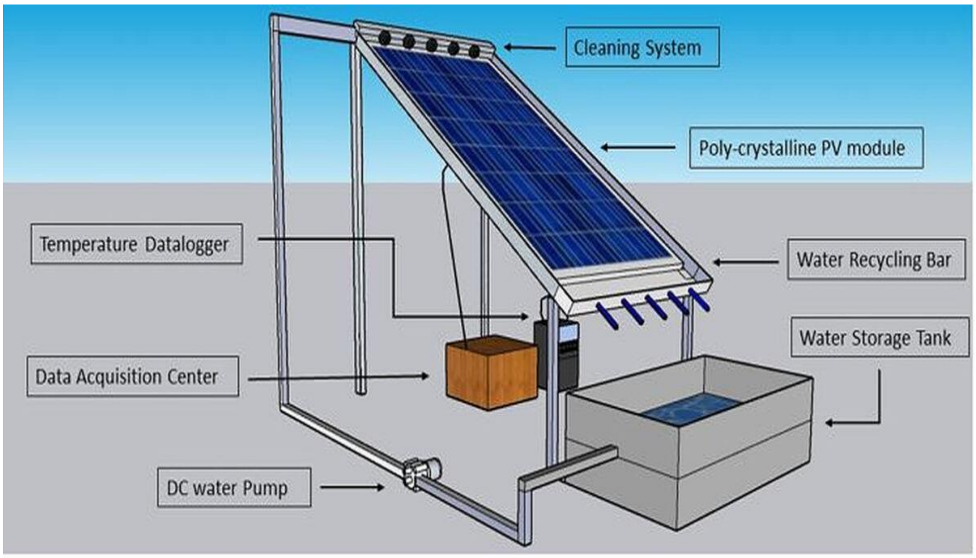


Fig 2.1: Water-Based Cleaning Systems

3. **Electrostatic Cleaning Systems**:

Electrostatic cleaning systems use electrostatic charges to attract and remove dust and debris from the surface of the solar panels. These systems can be highly effective at removing fine dust particles, but they require a power source to generate the electrostatic charge.

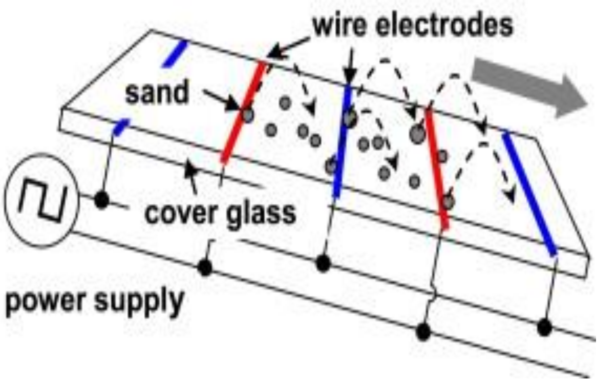


Fig 2.2: Electrostatic Cleaning System

4. **Self-Cleaning Coatings**:

Self-cleaning coatings are applied to the surface of the solar panels to prevent dust and dirt from adhering to the surface. These coatings use hydrophobic or superhydrophobic materials to repel water and dirt, allowing them to be easily washed away by rain or dew. Self-cleaning coatings can reduce the frequency of manual cleaning and maintenance, improving the overall efficiency of the solar panels.

Overall, automatic dust cleaning systems offer a convenient and effective way to maintain the efficiency and performance of solar panels. By removing dust and debris from the panel's surface, these systems can help maximize energy production and reduce maintenance costs over time.

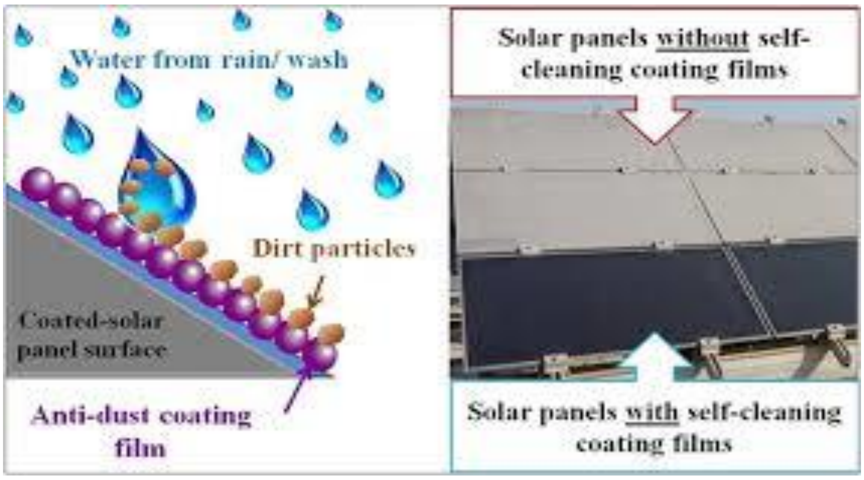


Fig 2.3: Self-Cleaning Coatings

2.3 VOLTAGE MONITORING TECHNIQUES IN SOLAR APPLICATION

Voltage monitoring is a critical aspect of solar energy systems, as it helps ensure the safe and efficient operation of the system. Monitoring the voltage of solar panels and batteries allows system operators to track the performance of the system, detect faults or anomalies, and optimize energy production and storage. Several techniques are commonly used for voltage monitoring in solar applications:

2.3.0 Direct Measurement:

Direct measurement involves connecting a voltmeter or multimeter directly to the terminals of the solar panel or battery to measure the voltage. This method provides a real-time reading of the voltage and is relatively simple and cost-effective. However, direct measurement requires physical access to the system and may not be suitable for remote monitoring.

2.3.1 Voltage Sensors:

Voltage sensors are electronic devices that can be used to measure the voltage of a solar panel or battery. These sensors are connected to the system and provide a digital or analog output proportional to the voltage. Voltage sensors can be integrated into a monitoring system to provide continuous monitoring and data logging capabilities.

2.3.2 Voltage divider network and microcontroller:

Using a voltage divider network with two resistors, the input voltage is scaled down to a level suitable for the microcontroller's analog-to-digital converter (ADC). The ADC converts this output voltage to a digital value, enabling the microcontroller to measure the voltage. This method is frequently employed in solar energy systems to monitor solar panel and battery voltages, providing crucial data for system optimization and performance tracking.

CHAPTER 3

SYSTEM DESIGN

3.0 BLOCK DIAGRAM

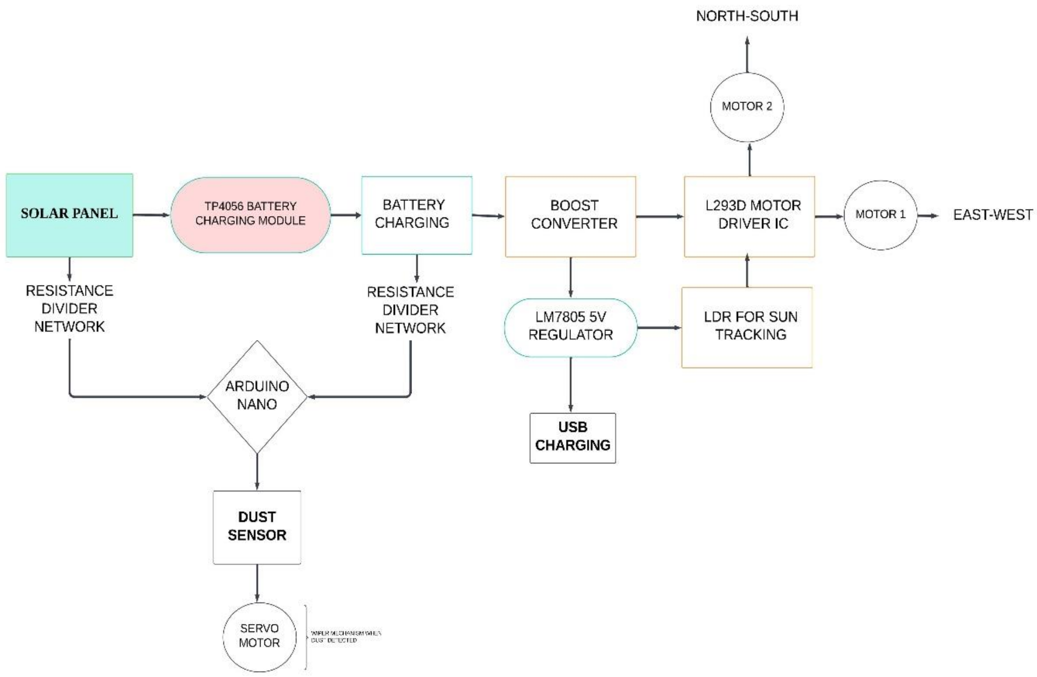


Fig 3.0: Block diagram of the project

3.1 EXPLANATION OF SYSTEM ARCHITECTURE

The system architecture of the dual-axis solar tracker with automatic dust cleaning and voltage monitoring comprises several key components that work together to achieve efficient solar energy generation and monitoring.

- **Solar Panels:**

  The solar panels are the primary energy harvesting components of the system. They convert sunlight into electrical energy, which is then used or stored for later use.

- **Dual-Axis Solar Tracker:**

  The dual-axis solar tracker is responsible for orienting the solar panels towards the sun to maximize energy capture. It uses an L293D motor IC and LDR to control the movement of the panels along both the east-west and north-south axes.

- **Automatic Dust Cleaning Mechanism:**

  The automatic dust cleaning mechanism consists of a dust sensor and a servo motor. The dust sensor detects the presence of dust on the solar panels, triggering the servo motor to activate a wiper mechanism that cleans the panels' surface.

- **Voltage Monitoring System:**

  The voltage monitoring system includes an Arduino Nano microcontroller, which is used to monitor the voltages of the solar panels and batteries. The Arduino Nano is connected to the panels and batteries through a voltage divider network, which scales down the voltages to levels suitable for the microcontroller's analog-to-digital converter (ADC).

- **Power Supply:**

  The system requires a stable power supply to operate efficiently. This can be provided by a battery.

- **Application of mobile charging**:

  The system uses USB module to charge the battery

3.2 COMPONENTS USED

Here is a list of components typically used in a dual-axis solar tracker with automatic dust cleaning and voltage monitoring:

- SOLAR PANELS (6V, 180mA)
- TP4056 BATTERY CHARGING MODULE
- LITHIUM ION BATTERY (3.7V ,1800mA)
- XL6009 DC-DC STEP UP MODULE
- LM7805 5V REGULATOR
- L293D MOTOR DRIVER IC
- LDR’S
- USB CHARGING MODULE
- BO MOTORS
- DUST SENSOR (GP2Y1014AU0F)
- SERVO MOTOR
- ARDUINO NANO
- RESISTORS
- CAPACITORS

- SWITCH

This is a general list, and the actual components used may vary based on the specific design and requirements of the solar tracker system.

1. **SOLAR PANEL:**

Solar panels are devices that convert sunlight into electricity through the PV effect. They are made up of solar cells, which are typically made from silicon and other materials. When sunlight hits the solar cells, it excites electrons, creating a flow of electric current.

We used two 6V, 180mA solar panels which are connected in parallel, their voltage remains the same (6V), but their current capacity is effectively doubled to 360mA. This configuration increases the overall power output of the solar panels, allowing for more energy to be harvested from the sunlight. Parallel connection is often used to increase the current capacity of a solar panel system without changing the voltage, making it suitable for applications where higher current is required.

Fig 3.1: Solar panels

2. **TP4056 BATTERY CHARGING MODULE:**

The TP4056 is a popular lithium-ion battery charging module that is widely used in DIY electronics projects and small electronic devices. It is designed to charge single-cell lithium-ion or lithium polymer batteries safely and efficiently.

The TP4056 module typically consists of a TP4056 charging chip, a few passive components (such as resistors and capacitors), and a micro USB port for power input. It also includes an LED indicator to show the charging status.

The module is very easy to use. You simply connect the positive and negative terminals of your battery to the corresponding terminals on the module, and then connect a 5V power source (such as a USB charger) to the micro USB port. The module will then charge the battery according to the specified charging parameters of the TP4056 chip.

One of the key features of the TP4056 module is its built-in protection functions, which help prevent overcharging, over-discharging, and short circuits, ensuring the safety of the battery during charging. Overall, the TP4056 module is a reliable and cost-effective solution for charging lithium-ion batteries in a variety of applications**.**

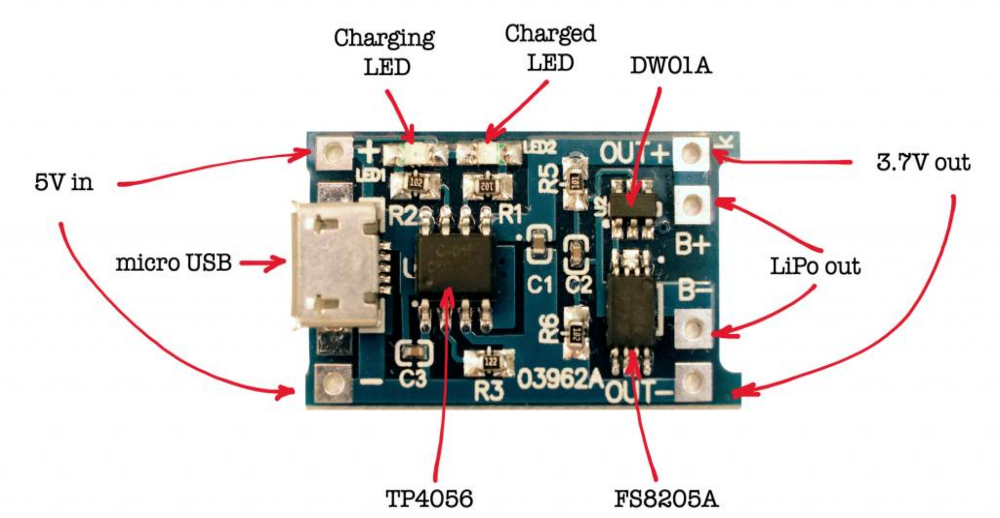


Fig 3.2: TP4056 Battery charging module

3. LITHIUM ION BATTERY:

Lithium-ion (Li-ion) batteries are a type of rechargeable battery commonly used in portable electronics, electric vehicles, and renewable energy systems. They are known for their high energy density, which allows them to store a large amount of energy in a relatively small and lightweight package. Li-ion batteries are also known for their high efficiency and long cycle life, making them ideal for applications where a rechargeable and long-lasting power source is required. However, they require careful handling and charging to prevent overheating, overcharging, and other safety issues.

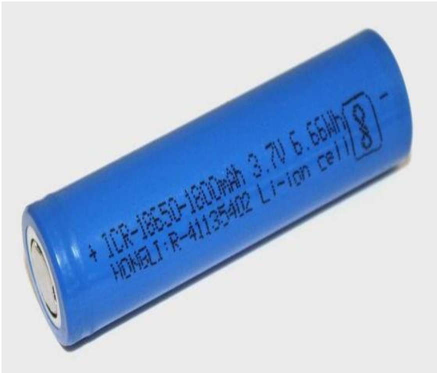


Fig 3.3: Lithium ion Battery

4. XL6009 DC-DC STEP UP MODULE:

The XL6009 is a DC-DC step-up (boost) module commonly used to convert lower DC voltages to higher DC voltages efficiently. Featuring an XL6009 adjustable boost converter IC, this module can boost input voltages as low as 5V up to output voltages as high as 32V, with a range of adjustable output voltages. Its design includes components like an inductor, capacitors, resistors, and a potentiometer for voltage adjustment. The XL6009 module is known for its high efficiency, achieved through synchronous rectification technology, which minimizes power loss and heat generation. These features make the XL6009 DC-DC step-up module a versatile and reliable choice for various electronics projects requiring voltage conversion.

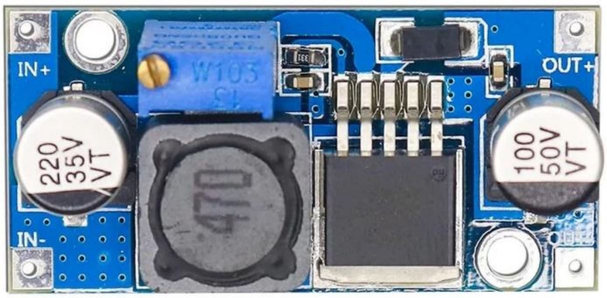


Fig 3.4: Xl6009 DC-DC Step up Converter

5. **LM7805 5V REGULATOR:** The LM7805 is a widely used linear voltage regulator that provides a stable 5V output voltage from a higher input voltage. It is commonly used to power low voltage electronic circuits from a higher voltage source.

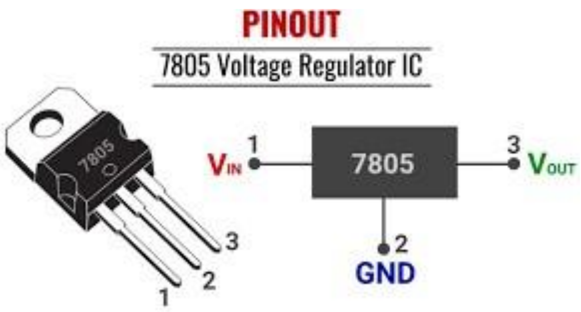


Fig 3.5: LM7805 5V Regulator

6. **L293D MOTOR DRIVER IC**: The L293D is a popular integrated circuit used to control the direction and speed of DC motors. It can drive two motors bidirectionally, making it ideal for robotic and motor control applications.

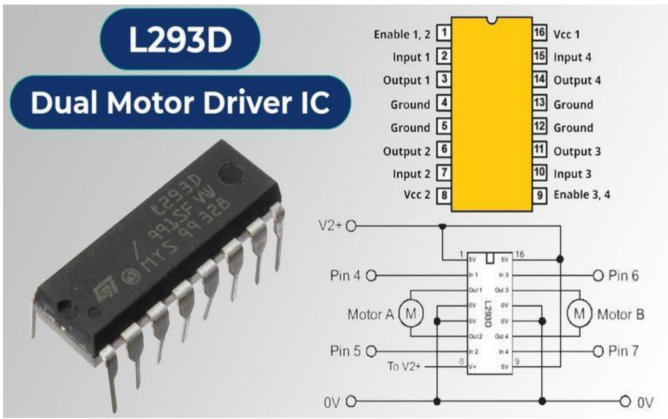


Fig 3.6: L293D Motor Driver IC

7. **LDR:** LDRs are light-sensitive resistors that change their resistance based on the amount of light falling on them. They are used in light sensing applications, such as automatic street lights and camera exposure control.

Fig 3.7: LDR

8. **USB CHARGING MODULE:** A USB charging module is a circuit that allows USB devices, such as smartphones and tablets, to be charged from a power source. It typically includes overcurrent protection and voltage regulation to ensure safe charging.

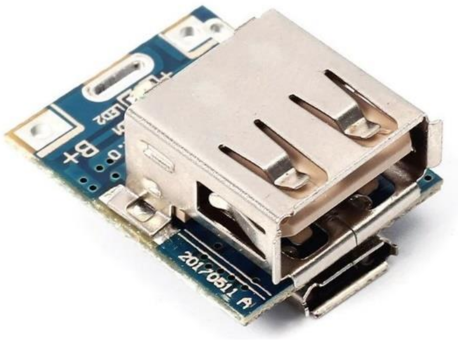


Fig 3.8: USB Charging Module

9. **BO MOTOR**: BO motors are a type of DC motor commonly used in robotics and DIY projects. They are known for their high torque and compact size, making them suitable for applications requiring precise control over movement.

Fig 3.9: BO Motor

10. **DUST SENSOR (GP2Y1014AU0F):**

    The dust sensor GP2Y1014AU0F is an optical sensor that uses an infrared LED and a phototransistor to detect the presence of dust particles in the air. It works on the principle of light scattering, where the light emitted by the LED is scattered by the dust particles. The phototransistor detects the scattered light, and the sensor calculates the dust concentration based on the intensity of the scattered light.

The GP2Y1014AU0F is suitable for detecting fine dust particles with a diameter of 0.5 micrometers or larger. It has a built-in amplifier circuit that converts the phototransistor's output into a voltage proportional to the dust concentration. This voltage can be read by a microcontroller, such as the Arduino Nano, for further processing and display.

One important consideration when using the GP2Y1014AU0F dust sensor is the need for periodic calibration, as dust accumulation on the sensor's optical components can affect its accuracy. Regular maintenance, such as cleaning the sensor's optical surfaces, can help maintain its performance over time.

Fig 3.10: Dust Sensor

11. SERVO MOTOR:

A servo motor is a type of motor that is capable of precise control of angular position. It is commonly used in robotics and remote control applications, such as steering mechanisms and robotic arms.

Fig 3.11: Servo Motor

12. **ARDUINO NANO:** The Arduino Nano is a small, breadboard-friendly microcontroller board based on the ATmega328P. It is compatible with the Arduino IDE and is used in a wide range of electronics projects.

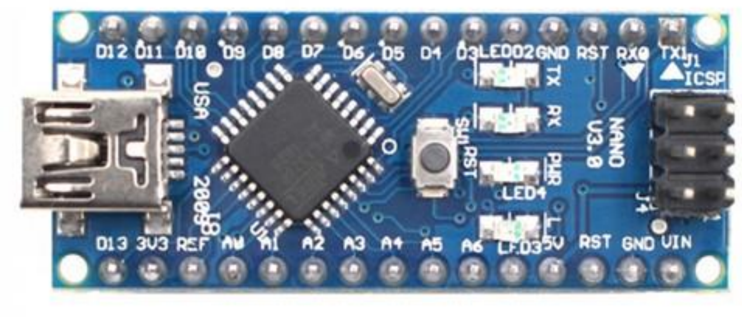


Fig 3.12: Arduino Nano

13. **RESISTOR:** Resistors are passive electronic components that limit the flow of electric current in a circuit. They are used to control the voltage and current in a circuit, and to divide voltages and currents. We have used 100k resistance for voltage divider network.
14. **CAPACITOR:** Capacitors are passive electronic components that store and release electrical energy. They are used in filtering, decoupling, timing, and energy storage applications in electronic circuits. We have used 220UF capacitance at lm7805 for filter purpose.
15. **SWITCH:** A switch is an electrical component that can make or break an electrical circuit. It is used to control the flow of current in a circuit, and can be used to turn devices on or off.

3.3 SCHEMATIC DIAGRAMS

3.3.0 SCHEMATIC DIAGRAM OF SOLAR AND BATTERY VOLTAGE MONITORING

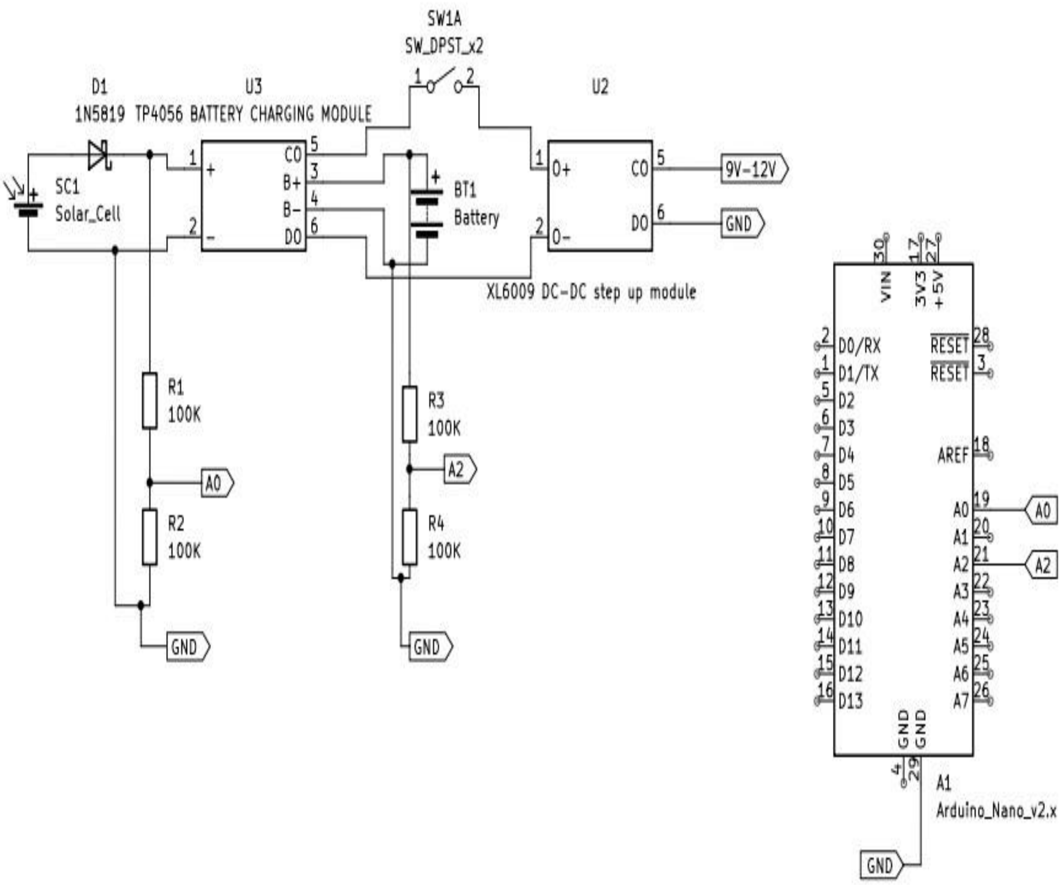


Fig 3.13: Schematic diagram of solar and battery voltage monitoring

3.3.1 SCHEMATIC DIAGRAM OF SOLAR SUN TRACKING AND MOBILE CHARGING

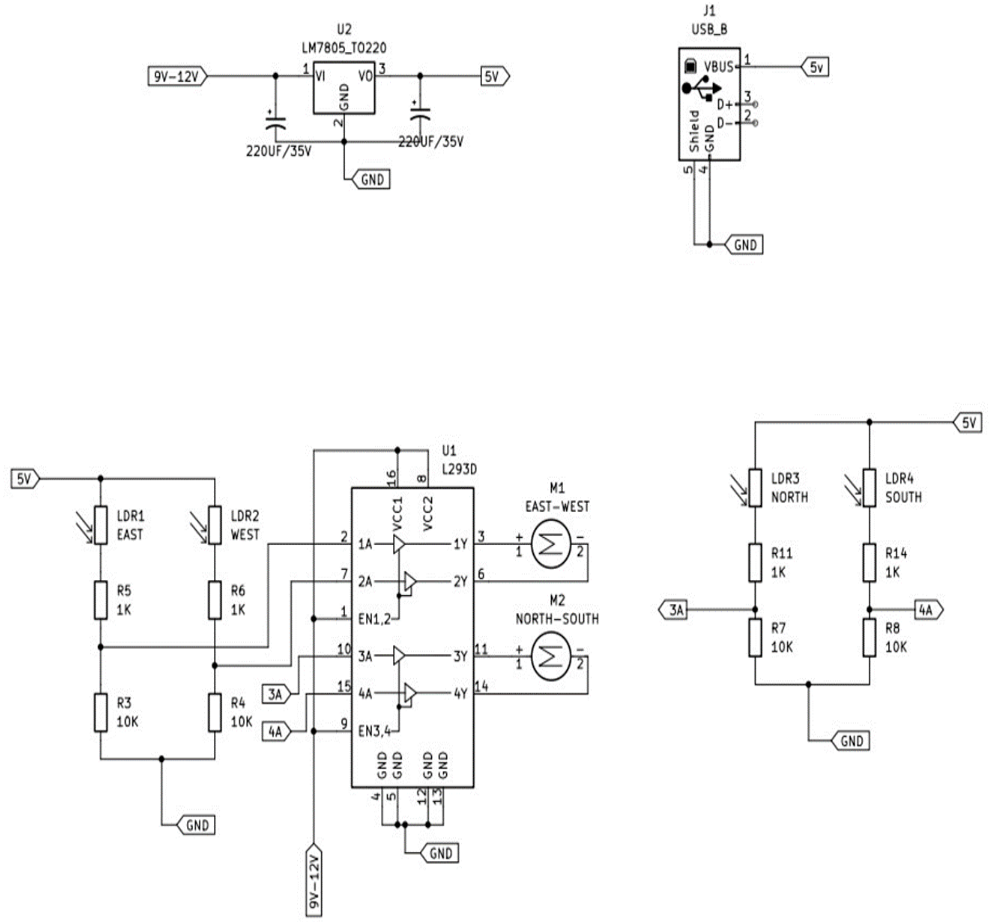


Fig 3.14: Schematic diagram of Sun tracking and mobile charging

3.3.2 SCHEMATIC DIAGRAM OF DUST SENSOR CONNECTED TO ARDUINO

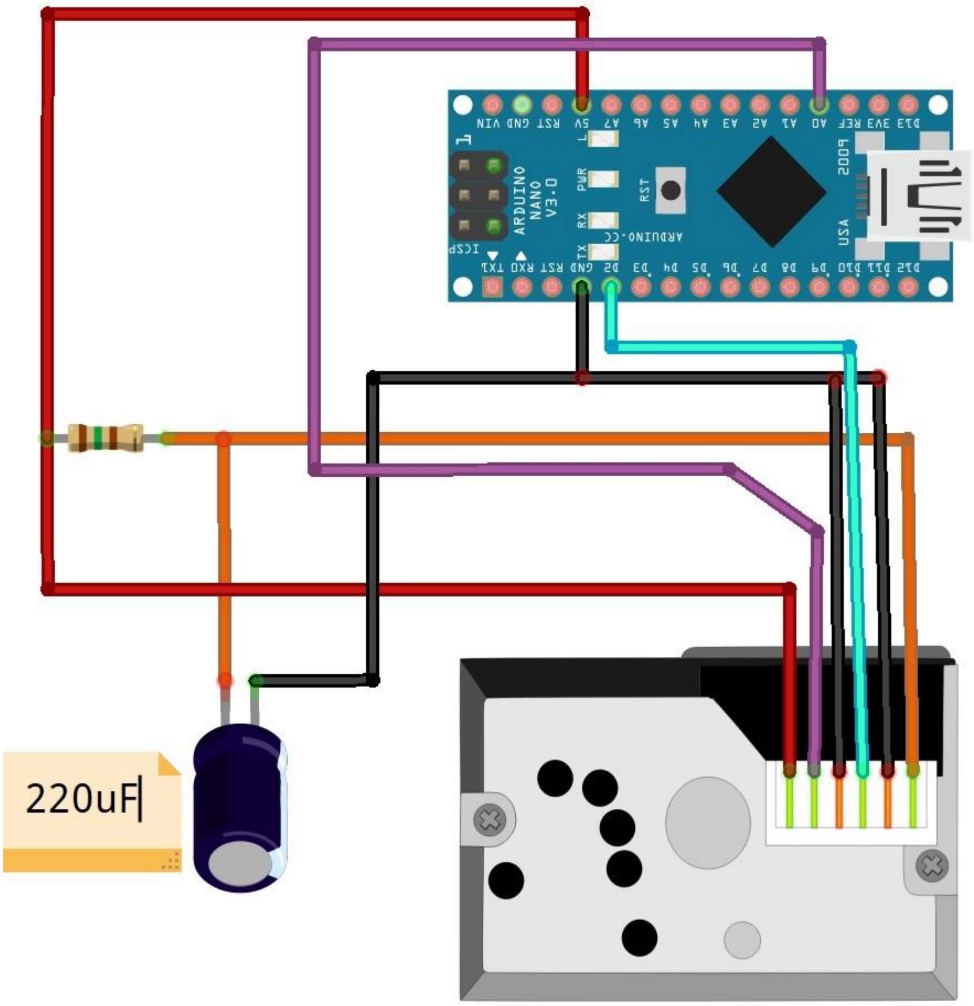


Fig 3.15: Schematic diagram of dust sensor connected to Arduino

3.4 HARDWARE IMPLEMENTATION

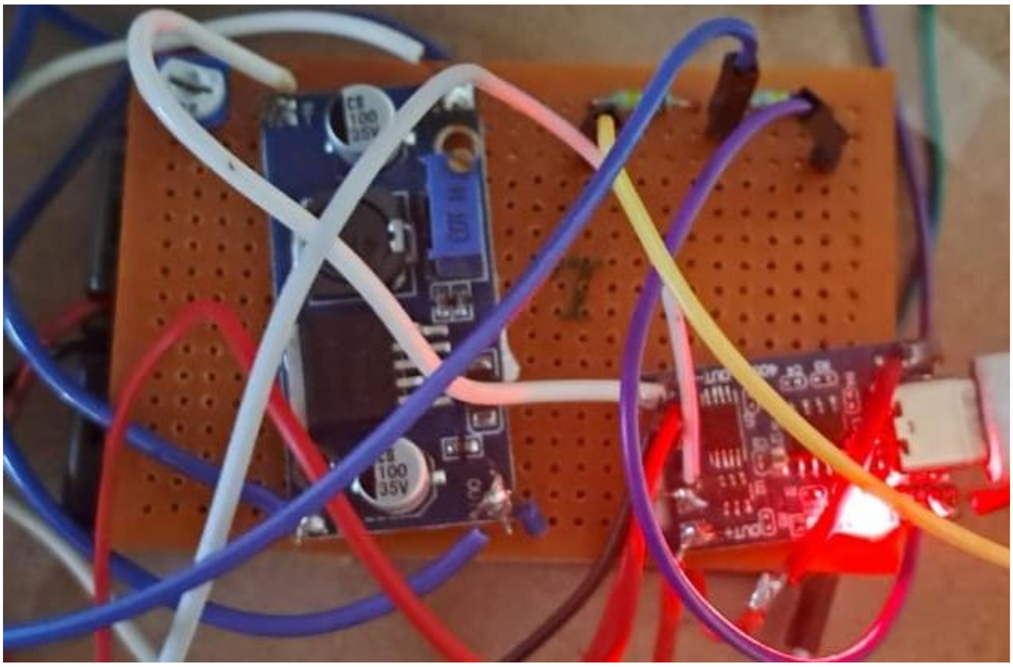


Fig 3.16: Hardware implementation of fig 3.13

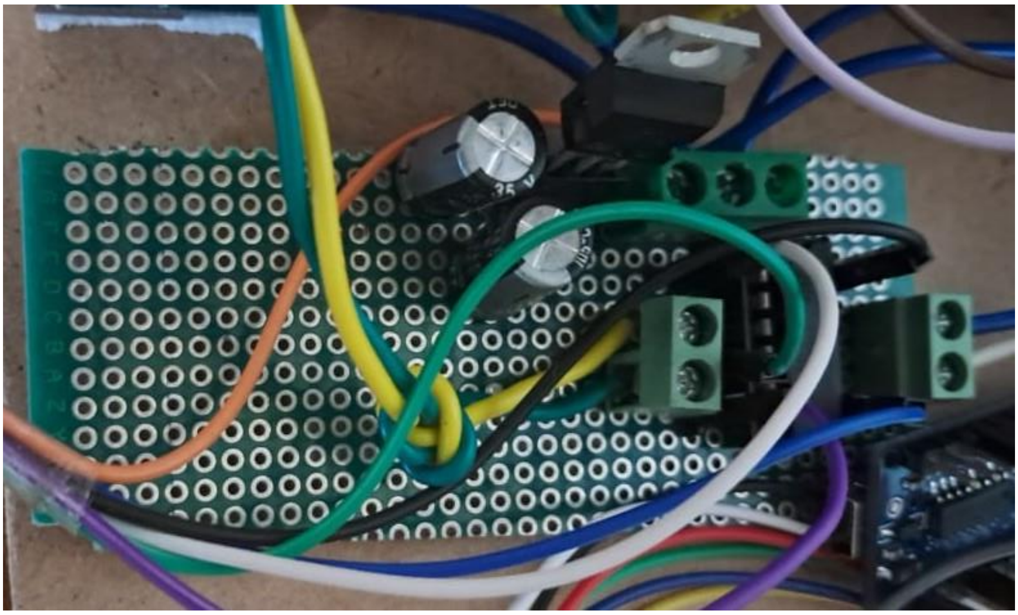


Fig 3.17: Hardware implementation of fig 3.14

3.5 WORKING PRINCIPLE OF L293D MOTOR IC FOR DUAL-AXIS TRACKING

The L293D is a popular motor driver IC commonly used in robotics and other applications requiring motor control. In the context of dual-axis solar tracking, the L293D is used to control the movement of the solar panels along both the east-west and north-south axes. The working principle of the L293D motor IC for dual-axis tracking can be explained as follows:

1. **H-Bridge Configuration:**

   The L293D IC contains two H-bridge circuits, each capable of driving a single motor bidirectionally. In the context of dual-axis tracking, one H-bridge can be used to control the motor responsible for east-west movement, while the other H-bridge can control the motor for north-south movement. The H-bridge configuration allows the motors to be driven forward or backward to move the solar panels in the desired direction.

2. **Control Signals:**

   The L293D IC requires four control signals to operate each H-bridge: two inputs for controlling the direction of the motor (e.g., forward or reverse) and two inputs for providing PWM (Pulse Width Modulation) signals to control the speed of the motor. By varying the PWM signal's duty cycle, the speed of the motor can be adjusted, allowing for precise control of the solar panel's movement.

3. **Logic Levels**:

   The L293D IC operates at TTL (Transistor-Transistor Logic) logic levels, meaning it can be directly interfaced with microcontrollers or other digital logic circuits. This makes it easy to integrate the L293D into a larger control system for the dual-axis solar tracker, allowing for automated tracking based on input from light sensors or other sources.

4. **Current Sensing**:

   The L293D IC includes built-in current sensing and thermal shutdown protection features. These features help protect the IC from damage due to overcurrent or overheating, ensuring reliable operation in demanding environments.

5. **Efficiency and Reliability:**

Overall, the L293D motor IC is known for its efficiency and reliability in motor control applications. Its H-bridge configuration, compatibility with TTL logic levels, and built-in protection features make it well-suited for use in dual-axis solar tracking systems, where precise and reliable motor control is essential for optimal performance.

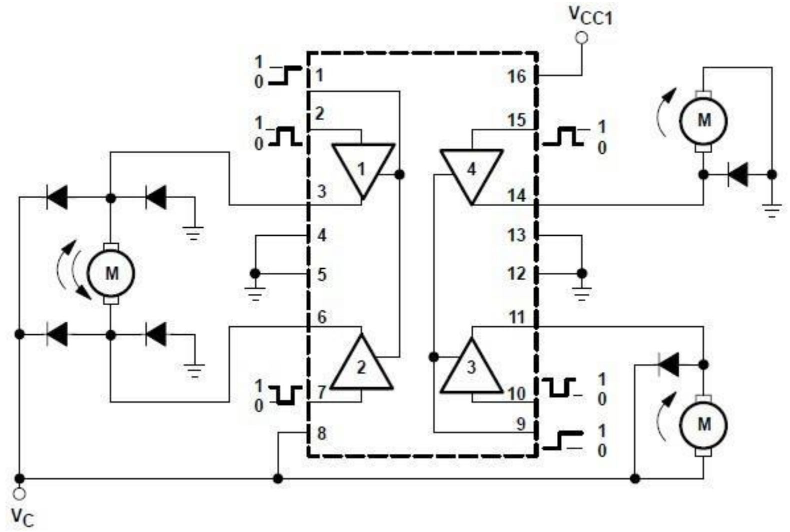


Fig 3.18: Schematic l293d motor IC for sun tracking

3.6 OPERATION OF SERVO MOTOR FOR DUST CLEANING

The operation of the servo motor for dust cleaning in the context of a dual-axis solar tracker involves using a dust sensor to detect the accumulation of dust on the solar panels and triggering the servo motor to actuate a cleaning mechanism when the dust level exceeds a certain threshold value.

1. **Dust Sensor Threshold Value:**

The dust sensor is calibrated to measure the amount of dust particles on the solar panels. When the dust level reaches a predefined threshold value, which can be set based on the sensitivity of the sensor and the desired cleaning frequency, the sensor sends a signal to the microcontroller indicating that cleaning is required.

2. **Microcontroller Control:**

   Upon receiving the signal from the dust sensor, the microcontroller activates the servo motor to engage the cleaning mechanism. The microcontroller may also record the time and date of the cleaning operation for monitoring and maintenance purposes.

3. **Servo Motor Activation:**

   The servo motor is connected to the cleaning mechanism, which could be a wiper or brush system, designed to remove dust from the surface of the solar panels. The servo motor is controlled by the microcontroller to move the cleaning mechanism across the surface of the panels, effectively removing the accumulated dust.

4. **Cleaning Cycle**:

   Once the cleaning cycle is complete, the servo motor returns the cleaning mechanism to its initial position, and the system goes back to monitoring the dust level. The process repeats whenever the dust sensor detects that the dust level has exceeded the threshold value, ensuring that the solar panels remain clean and efficient in capturing sunlight for energy generation.

By utilizing a dust sensor and servo motor in this way, the dual-axis solar tracker can maintain optimal performance by regularly cleaning the solar panels and ensuring maximum energy production efficiency.

3.7 VOLTAGE MONITORING CIRCUIT DESIGN

The voltage monitoring circuit design using a resistor dividing network and Arduino Nano involves creating a circuit that can measure the voltage of a solar panel or battery and convert it to a level suitable for the Arduino Nano's analog-to-digital converter (ADC). Here's how the circuit can be designed:

1. **Resistor Dividing Network:**

   The resistor dividing network consists of two resistors connected in series between the positive and negative terminals of the solar panel or battery. The junction between the two resistors is connected to the Arduino Nano's analog input pin.

2. **Calculation of Resistor Values:**

   The values of the two resistors in the divider network are chosen based on the input voltage range and the ADC's reference voltage. The output voltage (Vout) of the divider network can be calculated using the voltage divider formula:

$$\mathbf{Vout = (R2/R1 + R2) * Vin}$$

   Where Vin is the input voltage, R1 is the resistance of the first resistor, R2 is the resistance of the second resistor, and Vout is the output voltage.

3. **Arduino Nano Configuration:**

   The Arduino Nano is programmed to read the voltage at the analog input pin using the ADC. The ADC converts the analog voltage to a digital value, which can be read by the Arduino Nano and used for further processing or display.

4. **Calibration:**

   To ensure accurate voltage measurements, the voltage divider network may need to be calibrated. This involves comparing the measured voltage with a known reference voltage and adjusting the resistor values as needed to match the reference voltage.

5. **Safety Considerations:**

   It is important to ensure that the resistor values are chosen such that the maximum voltage across them does not exceed their power rating. Additionally, the voltage divider network should be protected from transient voltages and other electrical hazards.

Overall, the voltage monitoring circuit using a resistor dividing network and Arduino Nano provides a simple and cost-effective way to measure the voltage of a solar panel or battery, making it ideal for use in solar energy systems and other electronics projects.

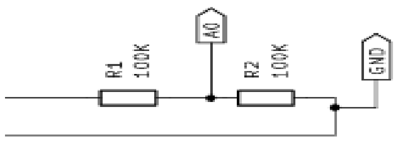


Fig 3.19: Voltage divider circuit

CHAPTER 4

IMPLEMENTATION

4.0 SOFTWARE DEVELOPMENT

The software development for the dual-axis solar tracker with voltage monitoring and automatic dust cleaning involves programming the Arduino Nano microcontroller to control the movement of the solar panels, monitor the voltage levels, and manage the dust cleaning mechanism. Here's an overview of the software development process:

1. **Movement Control**:

   The Arduino Nano is programmed to control the movement of the solar panels along both the east-west and north-south axes using the L293D motor IC. The program uses pulse-width modulation (PWM) signals to adjust the speed and direction of the motors based on input from light sensors or a predetermined tracking algorithm.

2. **Voltage Monitoring**:

   The Arduino Nano reads the voltage levels of the solar panels and batteries using the voltage divider network and the ADC. The program can then perform calculations or make decisions based on the voltage readings, such as adjusting the charging rate of the batteries or sending an alert if the voltage drops below a certain threshold.

3. **Automatic Dust Cleaning:**

   When the dust sensor detects a high level of dust on the solar panels, the Arduino Nano activates the servo motor to engage the cleaning mechanism. The program controls the servo motor to move the cleaning mechanism across the surface of the panels, ensuring effective cleaning.

4. **User Interface:**

   The software can include a user interface, such as an LCD display or a web-based interface, to provide real-time monitoring of the system's performance. The interface can display voltage levels, tracking status, and cleaning operations, allowing users to monitor and control the system easily.

Overall, the software development for the dual-axis solar tracker involves writing code to control the movement of the solar panels, monitor the voltage levels, and manage the dust cleaning mechanism, ensuring efficient and reliable operation of the system.

4.1 CALIBRATION AND TESTING PROCEDURE

Calibration and testing are crucial steps in ensuring the accuracy, reliability, and optimal performance of a dual-axis solar tracker with voltage monitoring and automatic dust cleaning. Here's an overview of the calibration and testing procedure for such a system:

1. **Calibration of Voltage Monitoring Circuit:**

   Start by calibrating the voltage monitoring circuit to ensure accurate voltage readings. This involves comparing the measured voltage with a known reference voltage and adjusting the resistor values in the voltage divider network if necessary.

2. **Calibration of Dust Sensor:**

   Calibrate the dust sensor to accurately detect the dust level on the solar panels. This may involve exposing the sensor to different levels of dust and recording its response to determine the threshold value for activating the cleaning mechanism.

3. **Motor Control Calibration:**

   Calibrate the motor control algorithm to ensure smooth and accurate movement of the solar panels. This may involve adjusting the PWM signals sent to the L293D motor IC based on the input from light sensors or a tracking algorithm.

4. **Testing Procedure:**

   Conduct a series of tests to verify the functionality of the system under various conditions. This includes testing the movement of the solar panels in response to changes in light intensity, monitoring the voltage levels of the solar panels and batteries, and testing the automatic dust cleaning mechanism.

5. **Performance Testing:**

   Evaluate the performance of the system by measuring its energy efficiency and tracking accuracy. This can be done by comparing the energy output of the system with and without the tracker, as well as comparing the actual position of the solar panels with the expected position based on the sun's position.

6. **Reliability Testing:**

   Test the reliability of the system by running it continuously for an extended period and monitoring for any issues or failures. This helps ensure that the system can operate reliably over the long term.

7. **User Interface Testing:**

   Test the user interface, if applicable, to ensure that it provides accurate and useful information to the user. This includes testing the display of voltage levels, tracking status, and cleaning operations.

By following these calibration and testing procedures, you can ensure that your dual-axis solar tracker with voltage monitoring and automatic dust cleaning is calibrated correctly, functions as expected, and meets the performance requirements for efficient solar energy generation.

4.2 INTEGRATION OF COMPONENTS

The integration of components in a dual-axis solar tracker with voltage monitoring and automatic dust cleaning involves connecting and coordinating the various components to ensure they work together seamlessly. Here's how the integration process typically unfolds:

1. **Physical Integration:**

   Start by physically integrating the components into a single system. This includes mounting the solar panels, motors, dust sensor, and other hardware onto a sturdy frame or structure. Ensure that the components are securely mounted and properly aligned for optimal performance.

2. **Electrical Integration:**

   Connect the electrical components together according to the system design. This includes connecting the motors to the L293D motor IC, connecting the dust sensor to the Arduino Nano, and connecting the voltage divider network to the ADC inputs of the Arduino Nano. Use appropriate wiring and connectors to ensure reliable electrical connections.

3. **Programming:**

   Write the necessary code for the Arduino Nano to control the motors, monitor the voltage levels, and activate the dust cleaning mechanism. Ensure that the code is well-structured, efficient, and capable of handling the various tasks required for the system to function correctly.

4. **Testing and Debugging**:

   Test the integrated system to ensure that all components are working correctly and that they are interacting as expected. Debug any issues that arise during testing, such as incorrect motor movement, inaccurate voltage readings, or malfunctioning dust sensor.

5. **Optimization:**

   Fine-tune the system for optimal performance. This may involve adjusting the tracking algorithm, calibrating the voltage monitoring circuit, or optimizing the cleaning mechanism to ensure effective dust removal.

6. **User Interface Integration:**

   If the system includes a user interface, integrate it with the rest of the system to provide users with real-time monitoring and control capabilities. Ensure that the user interface is intuitive and provides relevant information about the system's performance.

7. **Safety Considerations:**

   Throughout the integration process, prioritize safety by ensuring that all components are properly insulated, grounded, and protected from environmental hazards. Test the system under various conditions to ensure its safety and reliability.

By carefully integrating the components of a dual-axis solar tracker with voltage monitoring and automatic dust cleaning, you can create a robust and efficient system for maximizing solar energy generation while minimizing maintenance requirements.

CHAPTER 5

RESULTS AND ANALYSIS

5.0 ANALYSIS

The analysis of the dual-axis solar tracker project involves evaluating its performance, efficiency, and effectiveness in achieving its goals. Here's a detailed analysis of the project:

1. **Performance Analysis:**

   The performance of the solar tracker can be analyzed based on its ability to accurately track the sun's movement and optimize the positioning of the solar panels for maximum energy generation. This can be measured by comparing the energy output of the solar panels with and without the tracker under different weather conditions and at different times of the day.

2. **Efficiency Analysis:**

   The efficiency of the solar tracker can be analyzed by comparing the energy output of the solar panels with the energy input required to operate the tracker, including the power consumption of the motors, sensors, and control electronics. A higher efficiency indicates that the tracker is effectively harnessing solar energy and minimizing energy losses.

3. **Reliability Analysis**:

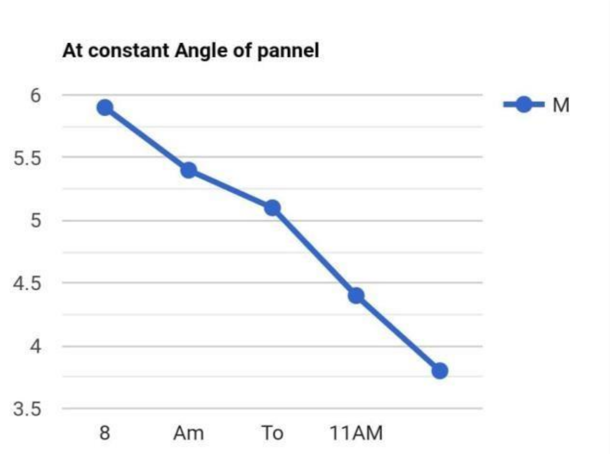
   The reliability of the solar tracker can be analyzed by monitoring its performance over an extended period and identifying any issues or failures that occur. This can help identify areas for improvement and ensure that the tracker can operate reliably over its expected lifespan.

Overall, a comprehensive analysis of the dual-axis solar tracker project can provide valuable insights into its performance, efficiency helping to guide future improvements and developments in solar energy technology.

Case:1 At Constant angle of Solar Panel (Without change in the position of the solar panel throughout the day)

Time	**Output Voltage**
8AMTO10AM	5.9
10 AM TO 12 PM	5.4
12PMTO2PM	5.1
2PMTO4PM	4.5
4PMTO6PM	3.8

TABLE 5.0 Time vs Output voltage without tracking

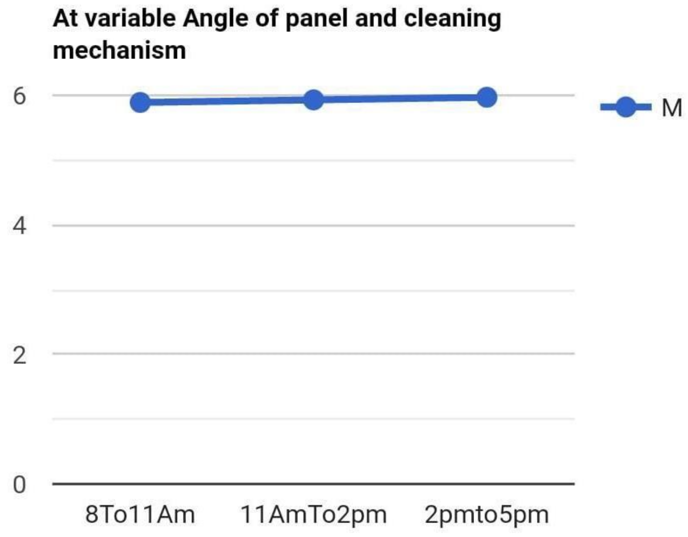


Graph 5.0: Graph representation of solar voltage without tracking

Case:2 At variable angle of panel and cleaning mechanism. (Change in position of thepanel w.r.t the sunlight throughout the day)

Panels Position	**Time**	**Output Voltage**
At initial position	8AMTO11AM	5.89
45 degree tilt from initialposition	12AM TO 2 PM	5.93
90 degree tilt from initialposition	3 PM TO 6 PM	5.97

TABLE 5.1 Time vs output voltage with tracking



Graph 5.1: Graph representation of solar voltage with tracking

5.1 RESULT

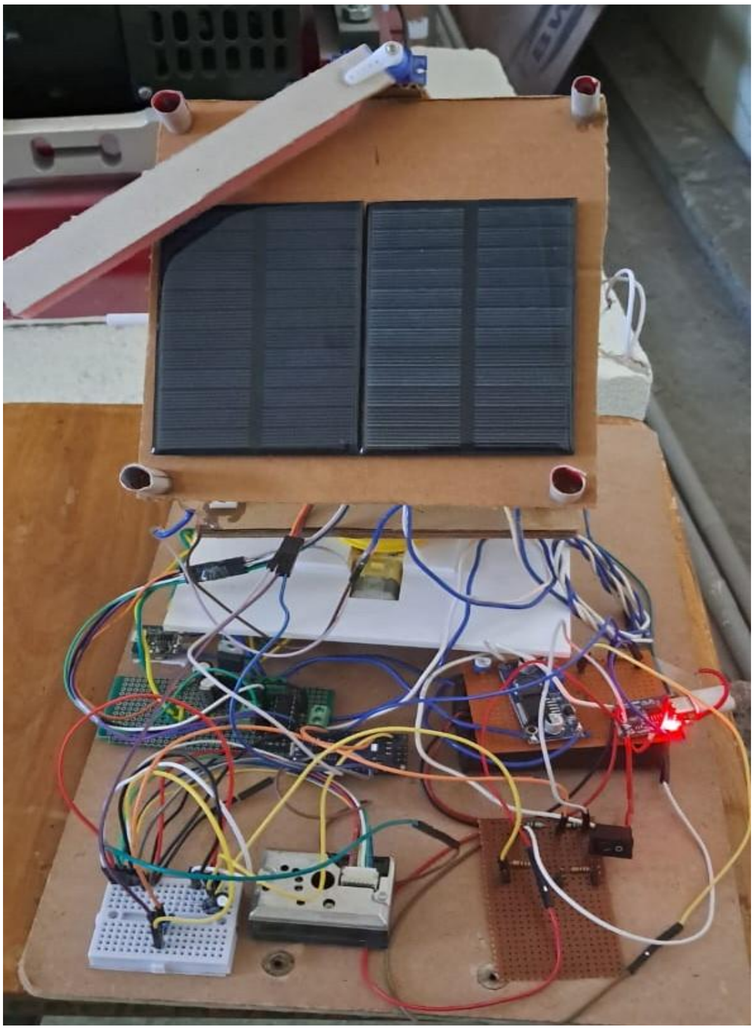


Fig 5.0: Prototype result

CHAPTER 6

CONCLUSION

In conclusion, the development of the dual-axis solar tracker with voltage monitoring and automatic dust cleaning represents a significant advancement in solar energy technology. By incorporating innovative features such as dual-axis tracking, voltage monitoring, and automatic dust cleaning, the project aims to maximize the efficiency and effectiveness of solar energy generation.

One of the key objectives of the project was to improve the energy output of solar panels by tracking the sun's movement along both the east-west and north-south axes. The dual-axis tracking capability allows the solar panels to maintain an optimal angle with respect to the sun throughout the day, maximizing the amount of sunlight captured and significantly increasing energy production.

Additionally, the inclusion of voltage monitoring ensures that the system operates within safe voltage limits, protecting the solar panels and batteries from damage due to overcharging or over-discharging. The automatic dust cleaning mechanism further enhances the system's efficiency by ensuring that the solar panels remain clean and free from dust, which can reduce their effectiveness in capturing sunlight.

Overall, the project has successfully demonstrated the feasibility and effectiveness of integrating advanced features into solar energy systems to improve their performance and efficiency. The dual-axis solar tracker with voltage monitoring and automatic dust cleaning has the potential to significantly impact the renewable energy sector by increasing the adoption and utilization of solar energy technology. Continued research and development in this field will further enhance the capabilities and efficiency of solar energy systems, making them a more viable and sustainable energy source for the future.

CHAPTER 7

FUTURE SCOPE

7.0 FUTURE SCOPE

- The future scope of the dual-axis solar tracker project with voltage monitoring and automatic dust cleaning is promising, with several avenues for further improvement and development. One key area for future enhancement is the integration of advanced tracking algorithms and sensors to improve the accuracy and efficiency of the solar tracking system. By using advanced algorithms and sensors, the tracker can adapt to changing environmental conditions in real-time, further optimizing the positioning of the solar panels for maximum energy generation.

- Another area for future development is the integration of smart grid technologies to enable the solar tracker to communicate with other devices and systems in the grid. This could allow the tracker to adjust its operation based on grid demand and availability, further increasing its efficiency and contribution to the overall grid stability.

- Furthermore, advancements in materials and manufacturing processes could lead to the development of more cost-effective and durable components for the solar tracker. This could reduce the overall cost of the system and make it more accessible to a wider range of users.

Overall, the future scope of the dual-axis solar tracker project is vast, with the potential to significantly improve the efficiency and effectiveness of solar energy systems. Continued research and development in this field will help drive innovation and lead to the widespread adoption of solar energy as a sustainable and reliable energy source.

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